Charts and summaries that tell the truth

Data engineering · Level 1 · Lesson 3



Today we will

- Choose a chart for the question.
- Work out the mean and the median, and see what one wrong value does.
- Draw a fair chart: bars from zero, a title and labels.

Remember

- Bars compare groups; lines show change.
- One wrong value drags the mean, not the median.
- Look at the chart before you trust a summary.

Our class chart

1. Warm-up (5 min): Which chart is fair?
2. Teach (8 min): Mean and median.
3. Draw (15 min): Our class chart.
4. Summarise (7 min): Mean, median and the odd one out.
5. Wrap-up (5 min): Which summary tells our story?

Prepare the 11, 12, 13, 14, 15 example on the board before the lesson. If your survey has a very large value, such as a pupil who travels for two hours, keep it: it's a good example of the median holding steady.

Let’s discuss

1. Why does one wrong value move the mean so much?
2. When would you show a line chart instead of bars?
3. What should every chart tell you, besides the bars?

Exit ticket

- Write five numbers whose mean and median are different, and say which one is fairer.